

Central retinopathy associated with ciliary body and choroidal melanomas

S E Skalicky, M Giblin, E C Figueira, R M Conway

Save Sight Institute, University of Sydney, NSW, Australia

Correspondence to:
Dr R M Conway, RANZCO,
Ocular Oncology Unit, c/- Save
Sight Institute, GPO Box 4337,
2001, NSW, Australia;
rmaxconway@hotmail.com

Accepted 3 September 2007

Various patterns of retinopathy can be responsible for the clinical presentation of uveal melanoma.¹ Occasionally a peripheral choroidal or ciliary body melanoma may present with macular pathology.² We present five cases of maculopathy due to peripheral or central choroidal melanomas. Cases 1–2 represent the first description, to our knowledge, of a peripheral uveal melanoma presenting with a premacular epiretinal membrane. In cases 3–5, the visual disturbance was due to foveal subretinal fluid, which has not been described previously in a preadolescent with ciliary body melanoma.

CASE REPORTS

Case 1 was a man aged 62, and cases 2–5 were women aged 58, 45, 9 and 38 years.

In cases 1 and 2, progressively impaired left visual acuity (6/18 and 6/15) led to the detection of large peripheral ciliochoroidal melanomas (fig 1). The first patient had no previous ocular pathology, and in particular had no retinal vascular occlusions, uveitis, trauma or retinal breaks. The second patient had been followed for many years with a previously stable epiretinal membrane, which had recently progressed causing new macular distortion. In both cases, unilateral left premacular membranes were separate from the tumour, associated with a small inferior dependent serous retinal detachment in case 1. There was no evidence of posterior vitreal detachment or other ocular pathology.

In cases 3–5, right subfoveal exudate caused acuities of 6/15, 6/36 and 6/9. In case 3, subretinal fluid was recognised 1 year prior to the emergence of a peripapillary melanoma and was initially diagnosed as idiopathic central serous retinopathy (CSR). The other patients had no previous ocular history. In case 4, a large superior ciliary body melanoma was separate from the serous detachment. In cases 3 and 5, the tumour was central (fig 2).

Treatment was with enucleation in cases 1–4 and transpupillary thermotherapy in case 5. Histopathology of resected specimens confirmed the diagnosis and revealed overlying retinal detachment in cases 1 and 3, and local retinal cystic degeneration in cases 3 and 4. No pathological evidence of generalised uveitis was seen apart from separate central serous detachment in case 4.

After a mean follow-up of 9 months (range 2–24) there has been no local or systemic recurrence.

COMMENT

Choroidal and ciliary body melanomas may cause overlying retinopathy including Bruch membrane

degeneration and cystic degeneration of the neural retina, and less commonly overlying serous detachment, direct invasion of the neural retina, central serous detachment, cystoid macula oedema and posterior uveitis.^{1,2} This series demonstrates a variable temporal and spatial link between retinopathy and the melanoma. In cases 2 and 3, retinopathy preceded clinical demonstration of the tumour by 12 months or more. In cases 1, 2 and 4, the maculopathy occurred with a separate, peripheral tumour.

Cases 1 and 2 describe an association between ciliochoroidal melanoma and premacular epiretinal gliosis, probably stimulated by inner limiting membrane disruption and vitreal exposure of the axonal layer. The ipsilateral nature of the maculopathy in an eye harbouring a uveal melanoma suggests that the association was not spurious. Potential mechanisms include low-grade uveitis with vitreal detachment or retinal traction, and the subretinal fluid in case 1 suggests a uveitic mechanism.³ However, the epiretinal membrane and the melanoma could have been present coincidentally.

Macular subretinal fluid caused by a peripheral melanoma may be due to dependent tracking of local fluid; anterior traction and vitreal shrinkage from tumour-induced uveitis; and nutritional insult in the context of choroidal vascular changes.^{2,4,5} Case 3 describes central fluid which preceded the tumour by 1 year. A case of idiopathic CSR preceding a central melanoma has been described once previously, when the posterior pole

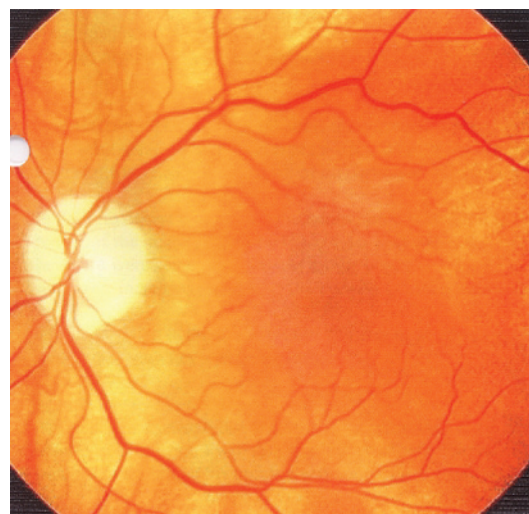


Figure 1 Epiretinal membrane at the fovea in an eye harbouring a large peripheral ciliochoroidal melanoma (case 1).

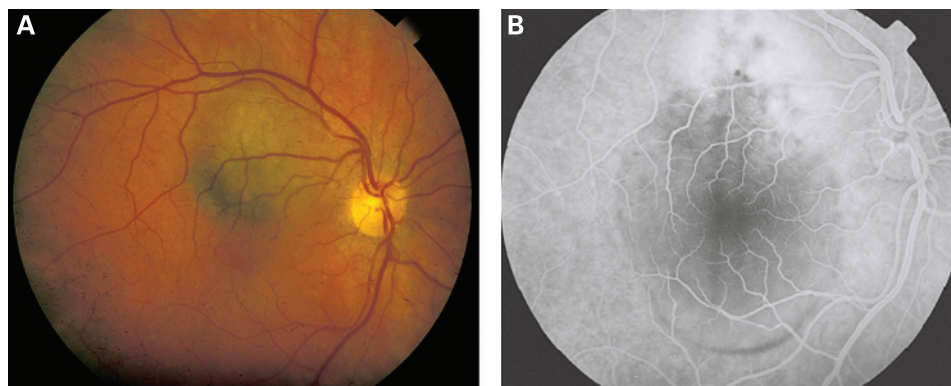


Figure 2 Central subretinal exudate: (A) foveal melanotic lesion with residual subretinal fluid and (B) corresponding fluorescein angiography (case 5).

melanoma emerged 9 years after the central fluid was first detected.⁵

These five cases demonstrate some of the patterns of central retinopathy associated with choroidal or ciliary body melanoma. They highlight the importance of dilated fundal examination and a low threshold for clinical suspicion of choroidal or ciliary body tumours.

Competing interests: None.

REFERENCES

1. **Damato BE**, Foulds WS. Tumour-associated retinal pigment epitheliopathy. *Eye* 1990;**4**:382–7.
2. **Wolter JR**. Three basic types of foveal involvement in choroidal melanomas. *Graefes Arch Clin Exp Ophthalmol* 1982;**218**:237–43.
3. **Ting FS**, Kwok AK. Treatment of epiretinal membrane: an update. *Hong Kong Med J* 2005;**11**:496–502.
4. **Wolter JR**. Foveal pathology in ciliary body melanoma. *Ophthalmic Surg* 1982;**13**:309–11.
5. **Reichel E**, Shore JW, Brucker AJ, *et al*. Occult choroidal melanoma presenting as a serous macular detachment. *Retina* 1995;**15**:167–9.